

Product Strategy & Architecture Plan

AI Learning Platform: Data Engineering Curriculum

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Location: Lake Forest, IL

Status: Architecture & Discovery Phase

Vision Statement: To build a scalable, intelligent learning application that breaks down complex technical concepts into atomic, gamified micro-lessons. The platform will utilize Google AI Studio to power a dynamic feedback loop, automatically adjusting lesson difficulty based on user comprehension.

1. The Story Map

The platform is divided into three core pillars: The Shell (User Interface), The Brain (AI Engine), and the initial Curriculum.

Part 1: The Platform (The Shell)

Activity	Step	User Stories (MVP)
Onboarding & Setup	Define the Goal	• Select macro-concept to establish destination. • Complete baseline assessment for accurate level placement.
The Dashboard	Track Progress	• View visual "skill tree" indicating mastery progress. • Identify specific micro-skills flagged as weak by the AI engine.
The Sandbox	Engage Content	• Interact via drag-and-drop, input, or code interfaces. • Request conceptual hints without revealing answers.

Part 2: The AI Engine (The Brain)

Phase	System Step	Logic & Prompt Constraints
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Evaluation Loop	Assess Input	Calculate failure point against perfect solution. Output strict JSON mapping <code>`is_correct`</code> , <code>`error_type`</code> , and <code>`concept_failed`</code> .
Feedback Loop	Intervene	Translate evaluation into targeted, conversational micro-lessons explaining only the failed atomic concept.
Generation Loop	Adjust Difficulty	Succeed: Generate Level + 1. Fail: Generate Level 0 (same difficulty, new context) targeting the weak point.

Part 3: Dimensional Modeling Curriculum (V1)

A structured 10-level progression mapping data engineering fundamentals. This serves as the initial context window for the AI Engine.

- **Level 1: The Atomic Core** - Categorizing Facts vs. Dimensions (Drag & Drop).
- **Level 2: Star Schema Basics** - Mapping Primary to Foreign Keys.
- **Level 3: Normalization** - Refactoring a flat table into a Snowflake Schema.
- **Level 4: The Grain** - Defining row-level granularity for a specific business question.
- **Level 5: Time Travel** - Implementing Slowly Changing Dimensions (SCD Type 1 vs 2).
- **Level 6: Enterprise View** - Designing Conformed Dimensions across multiple facts.
- **Level 7: Events Without Metrics** - Modeling Factless Fact Tables (e.g., system logs).
- **Level 8: Many-to-Many** - Constructing Bridge Tables to resolve complex relationships.
- **Level 9: The Reality Check** - Handling late-arriving dimensions and orphan facts.
- **Level 10: End-to-End** - Designing a full logical model from a raw JSON payload.

2. Technical Architecture & Infrastructure

A cloud-native approach utilizing Google Cloud Platform (GCP) to ensure security and prevent "scalability nightmares." The infrastructure is designed to integrate smoothly with modern Python tooling (e.g., uv for backend dependencies) and broader data engineering ecosystems (like Azure Data Factory integrations in the future).

- **Frontend Application:** React/Next.js or Flutter for cross-platform delivery.
- **Secure Middleware:** Google Cloud Run or Firebase Cloud Functions. The client never communicates with AI Studio directly.
- **AI Service:** Google AI Studio API handling the Evaluation, Feedback, and Generation loops.
- **State Database:** Cloud Firestore for low-latency state management and progression tracking.

- **Telemetry & Analytics:** BigQuery. Prompt evaluations and user success rates are routed here to act as an ETL pipeline for continuous prompt refinement.

3. CI/CD & Engineering Standards

Strict deployment pipelines to ensure the non-deterministic nature of LLMs does not break application logic.

- **Environment Segregation:** Strict separation of Local Dev, Staging, and Production environments.
- **Secrets Management:** Google Cloud Secret Manager handles all API keys at runtime. No `.env` files are ever committed.
- **Deployment Pipeline (GitHub Actions):** Automated linting and unit testing on PRs. Merges trigger Staging deployments. Tagged releases deploy to Production.
- **Mock Testing:** Unit tests utilize mock JSON payloads to simulate Google AI Studio responses, ensuring system stability without incurring API costs during CI runs.

4. Project Roadmap: Next Actions

The following epics represent the immediate sequence of execution for the project.

Big Rock 1: The Engine Room (AI Studio Setup)

Draft system instructions and establish the strict structured output (JSON Schemas) for the Evaluation Loop. Build Level 1 Few-Shot prompts to prove the engine can accurately grade Fact vs. Dimension tasks.

Big Rock 2: Infrastructure Foundation

Initialize the GCP/Firebase project. Deploy the first secure Cloud Function to act as middleware. Configure Google Cloud Secret Manager for the AI Studio API keys.

Big Rock 3: The Evaluation Pipeline

Establish the data flow: Extract user payload from frontend, Transform via Cloud Function to AI Studio, and Load the resulting JSON assessment into Firestore to update user state.

Big Rock 4: CI/CD Guardrails

Configure GitHub Actions for automated testing. Implement mock frameworks to test the application's handling of AI schemas.

Big Rock 5: The Sandbox UI

Develop the interactive frontend components (drag-and-drop buckets, connection lines) and wire them to the secure Cloud Function endpoints.